

Roll No.

TPE-501

**B. TECH. (PE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
PETROLEUM GEOPHYSICS**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) What is Geophysics ? How does it play a major role in Petroleum Industry ? What are the various earth properties measured in Geophysics ? 10 Marks (CO1)

OR

- (b) Compare Geophysics with Geology. Define the various earth properties measured by Geophysical method. How are these properties used in defining the subsurface structure ? 10 Marks (CO1)
2. (a) How many geophysical methods are available for oil exploration ? Explain the Gravity Survey method in detail. 10 Marks (CO2)

OR

- (b) Explain passive and active method of geophysics. What is the objective of geophysical investigations ? 10 Marks (CO2)
3. (a) Explain the principle of magnetic survey method in details. Name the natural magnetic elements and ferromagnetic minerals. 10 Marks (CO2)

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OR

- (b) What is Electromagnetic method of geophysical survey ? Draw a neat diagram to explain it. Where is this survey useful ? 10 Marks (CO2)
4. (a) Explain the seismic method for exploration. What are the basic properties measured in this method ? 10 Marks (CO1, CO2)

OR

- (b) Explain P-Wave and S-wave along with figure. Write the formula for P-wave and S-wave velocity in terms of bulk modulus and shear modulus. 10 Marks (CO1, CO2)
5. (a) What is Huygens principle for wave propagation ? Draw a diagram for wave propagation and refraction. 10 Marks (CO2)

OR

- (b) Explain Snell's Law of reflection and refraction. What are acoustic impedance and reflection coefficients ? Write their formulas. 10 Marks (CO2)